Business Intelligence Systems

INFO 4555 / 5556

Fall 2025

Bottom-Up Dimensional Data Modeling approach



Bottom-up Dimensional Data Modeling slides

The content of this slide deck are from material created by:

Neil Belcher, Senior Data Modeler, Cornell University

Neil gave guest lectures on this topic from 2016 – 2020. Neil is unable to give his guest lecture this year. I feel the content is extremely relevant to this course, so will do my best to cover Neil's topic.

Requirements are Hard

Kimball strongly recommends the development of an enterprise data warehouse bus matrix to capture the relationships between the core business processes / events and core descriptive dimensions BEFORE development begins.

https://decisionworks.com/2003/09/the-bottom-up-misnomer/

Requirements are Hard

Between 70% to 80% of corporate business intelligence projects fail, according to research by analyst firm Gartner.

A combination of poor communication between IT and the business, the failure to ask the right questions or to think about the real needs of the business, means most business intelligence (BI) projects fail to deliver, the firm says.

https://www.computerweekly.com/news/1280094776/Poor-communication-to-blame-for-business-intelligence-failure-says-Gartner

Requirements are Hard

Taking the Enterprise View can be very daunting, very political and very time consuming.

Requirements are Hard

Look At Cornell As An Enterprise:

Agriculture and Life Sciences
Alumni Affairs and Development
Architecture Art and Planning
Arts and Sciences
Budget and Planning
College of Business
Computing and Information Science
Continuing Education and Summer Sessions
Cooperative Extension Association
Cornell Tech
Engineering
Financial Affairs
Graduate School
Human Ecology
Human Resources
Industrial & Labor Relations
Information Technology
Infrastructure, Properties and Planning
Institutional
Law School
President
Provost
Research Division
Student and Academic Services
Student and Campus Life
University Communications
University Library
University Relations
Veterinary Medicine
WCMC NY
WCMC QATAR
WMC ITHACA

Question

How do you balance the need to gather requirements vs. getting something built?



Question

You have to start somewhere.



Question

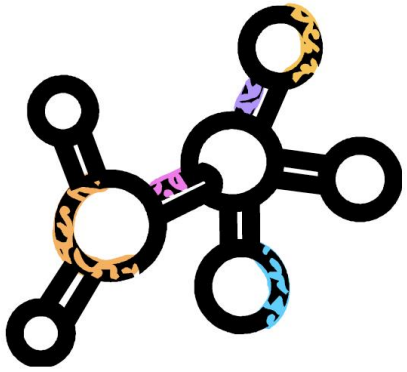
Do you really need to have user requirements in order to design a dimensional data model?



Answer

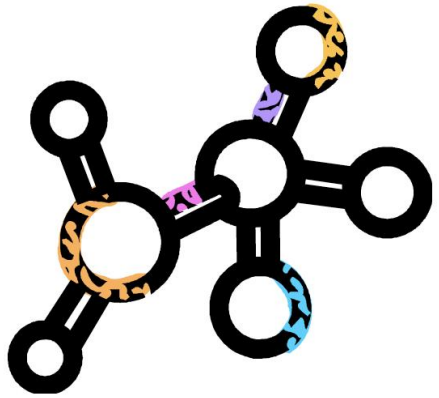
Not Really

Let the structure of the source data drive the design of the dimensional models.



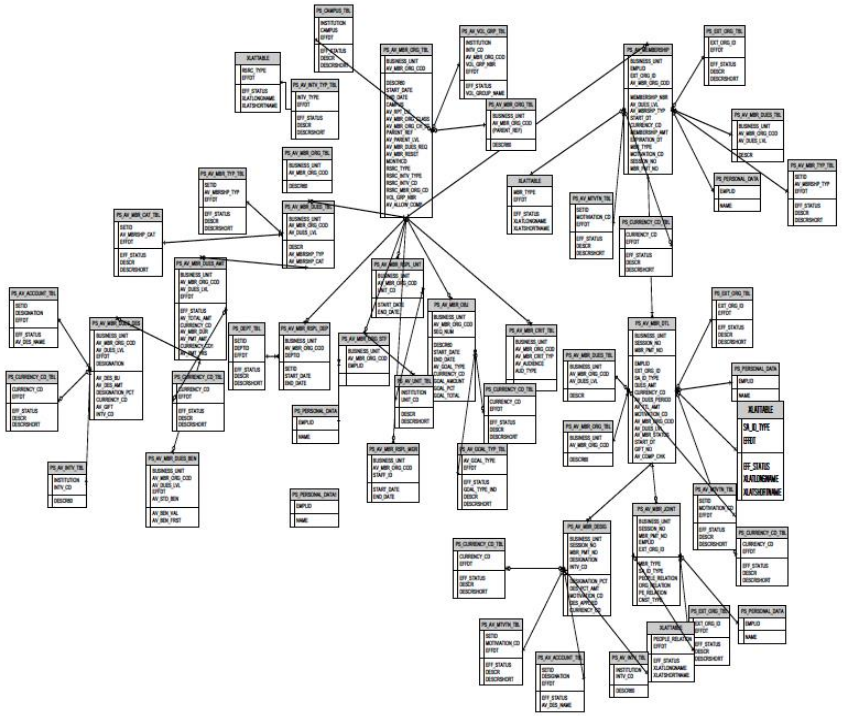
Answer

All the facts and dimensions **must** be in the source data, you just need to find them.



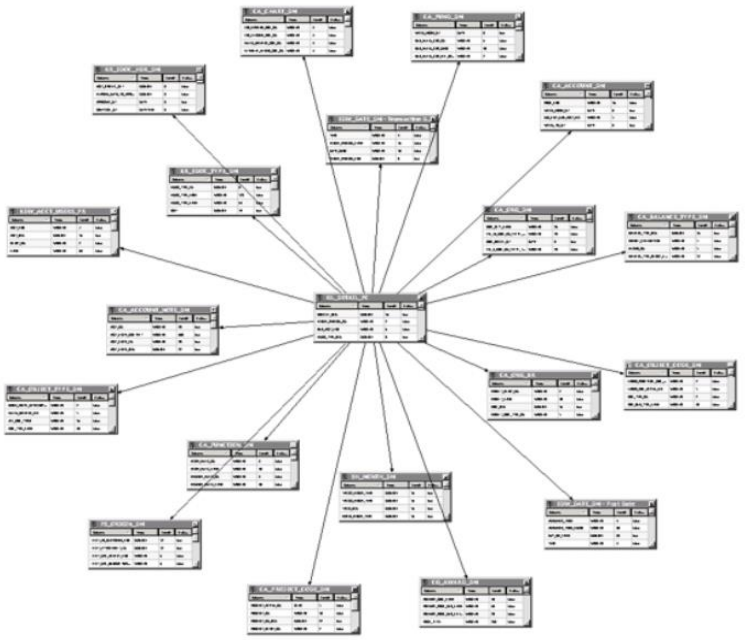
Your Mission as a Dimensional Data Modeler

SOURCE



ETL

TARGET



Traditional Top-Down Approach

- Gather User Requirements
- Develop Conceptual Models
- Develop Logical Models
- Develop Physical Models
- Assumes that users know their data and what their requirements are
- Data Structures inform the design late in the process

Bottom-Up Approach

- Start By Analyzing The Source Data
- Find A Foundational Fact
- Use The Primary Keys To Lead You To Obvious Dimensions
- Prototype The Dimensions And Facts (SQL)
- Test Your Prototype Model
- Enhance Models Based Upon Top-Down Requirements When They Become Available

The Bottom-Up Premise

Every data source has foundational facts and obvious dimensions that are irrefutable.



Just the Facts

- Foundational Facts Are Numbers: Dollars, Counts, Volumes, Units Sold, etc.
- Your Challenge Is To Identify The Most Obvious And Important Fact
- This Is The Starting Point For The Rest Of The Model(s)

- How Do You Find Facts?
  - Detective work
  - Metadata

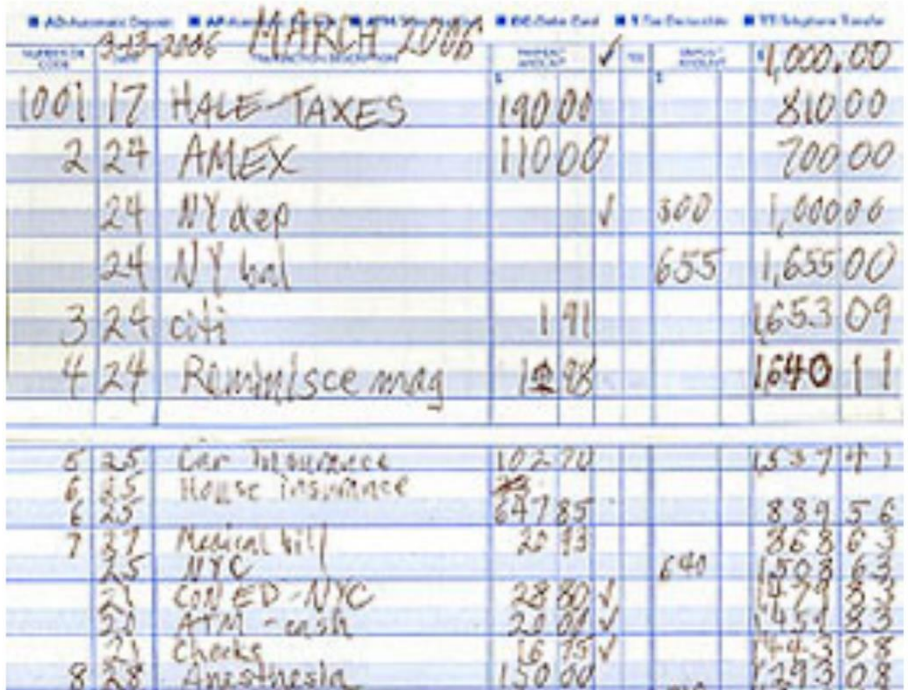
Just the Facts

Mission: Build A Financial Data Model For Cornell
Which Unit Tracks The $$$?

Agriculture and Life Sciences
Alumni Affairs and Development
Architecture Art and Planning
Arts and Sciences
Budget and Planning
College of Business
Computing and Information Science
Continuing Education and Summer Sessions
Cooperative Extension Association
Cornell Tech
Engineering
Financial Affairs
Graduate School
Human Ecology
Human Resources
Industrial & Labor Relations
Information Technology
Infrastructure, Properties and Planning
Institutional
Law School
President
Provost
Research Division
Student and Academic Services
Student and Campus Life
University Communications
University Library
University Relations
Veterinary Medicine
WCMC NY
WCMC QATAR
WMC ITHACA

Just the Facts

- Financial Affairs is the central unit that controls the Kuali Financial System (KFS) - Cornell's Check Book



Just the Facts

- All Cornell Units use KFS
  - Income – Tuition, Grants, Gifts, etc.
  - Expense – Salaries, Purchasing, Capital Projects, etc.
  - Assets, Liabilities as well.
- A perfect place to find a foundational fact!
- Find the table and column that records each Cornell financial transaction

Metadata

Metadata is "data [information] that provides information about other data".[1] Two types of metadata exist: **structural metadata** and **descriptive metadata**. Structural metadata is data about the containers of data. Descriptive metadata uses individual instances of application data or the data content.

(From Wikipedia)

Metadata

- Oracle Structural Metadata Views

```
SELECT *
 FROM all_views
WHERE view_name LIKE 'ALL%';
```

  - ALL_TABLES
  - ALL_TAB_COLUMNS
  - ALL_VIEWS
  - ALL_INDEXES
  - ALL_CONSTRAINTS …

Metadata

```
SELECT atc.table_name,
       atc.column_name,
       atc.data_type,
       atc.data_precision,
       atc.data_scale,
       atc.column_id,
       CASE WHEN pk.column_name IS NULL THEN 'N' ELSE 'Y' END
is_primary_key
  FROM all_tab_columns atc
       LEFT OUTER JOIN (SELECT DISTINCT cn.table_name,
                                        cl.column_name
                        FROM all_constraints cn,
                             all_cons_columns cl
                       WHERE cn.owner = 'KFS' AND cn.constraint_type = 'P'
                       AND cn.table_name = cl.table_name ) pk
         ON (atc.table_name = pk.table_name
         AND atc.column_name = pk.column_name)
 WHERE atc.owner = 'KFS'
ORDER BY 1, 6;
```

Metadata query result set

	A	B	C	D	E	F	G
1	TABLE_NAME	COLUMN_NAME	DATA_TYPE	DATA_PRECISION	DATA_SCALE	COLUMN_ID	IS_PRIMARY_KEY
2	AP_AUTO_APRV_EXCL_T	FIN_COA_CD	VARCHAR2			1	Y
3	AP_AUTO_APRV_EXCL_T	ACCOUNT_NBR	VARCHAR2			2	Y
4	AP_AUTO_APRV_EXCL_T	OBJ_ID	VARCHAR2			3	Y
5	AP_AUTO_APRV_EXCL_T	VER_NBR	NUMBER	8	0	4	Y
6	AP_AUTO_APRV_EXCL_T	DOBJ_MAINT_CD_ACTV_IND	VARCHAR2			5	Y
7	AP_CRDT_MEMO_ACCT_CHG_T	CRDT_MEMO_ACCT_CHG_ID	NUMBER	10	0	1	Y
8	AP_CRDT_MEMO_ACCT_CHG_T	OBJ_ID	VARCHAR2			2	Y
9	AP_CRDT_MEMO_ACCT_CHG_T	VER_NBR	NUMBER	8	0	3	Y
10	AP_CRDT_MEMO_ACCT_CHG_T	CRDT_MEMO_ITM_ID	NUMBER	10	0	4	N
11	AP_CRDT_MEMO_ACCT_CHG_T	FIN_COA_CD	VARCHAR2			5	Y
12	AP_CRDT_MEMO_ACCT_CHG_T	ACCOUNT_NBR	VARCHAR2			6	Y
13	AP_CRDT_MEMO_ACCT_CHG_T	SUB_ACCT_NBR	VARCHAR2			7	N
14	AP_CRDT_MEMO_ACCT_CHG_T	FIN_OBJECT_CD	VARCHAR2			8	Y
15	AP_CRDT_MEMO_ACCT_CHG_T	FIN_SUB_OBJ_CD	VARCHAR2			9	N
16	AP_CRDT_MEMO_ACCT_CHG_T	PROJECT_CD	VARCHAR2			10	N
17	AP_CRDT_MEMO_ACCT_CHG_T	ORG_REFERENCE_ID	VARCHAR2			11	N
18	AP_CRDT_MEMO_ACCT_CHG_T	ITM_ACCT_TOT_AMT	NUMBER	19	2	12	Y
19	AP_CRDT_MEMO_ACCT_CHG_T	ACLN_PCT	NUMBER	35	20	13	Y
20	AP_CRDT_MEMO_ACCT_CHG_T	CRDT_MEMO_ACCT_CHG_TS	DATE			14	Y
21	AP_CRDT_MEMO_ACCT_CHG_T	FDOC_POST_YR	NUMBER	4	0	15	N
22	AP_CRDT_MEMO_ACCT_CHG_T	FDOC_POST_PRD_CD	VARCHAR2			16	N
23	AP_CRDT_MEMO_ACCT_T	CRDT_MEMO_ACCT_ID	NUMBER	10	0	1	Y
24	AP_CRDT_MEMO_ACCT_T	OBJ_ID	VARCHAR2			2	Y
25	AP_CRDT_MEMO_ACCT_T	VER_NBR	NUMBER	8	0	3	Y

The Hunt for the Facts

- Looking for a needle in a haystack (out of 6800+ possible columns)
- How can you eliminate tables from consideration?
- You must become an expert at understanding the source tables/columns

The Hunt for the Facts

Kuali Financial System - KFS

- What is the table naming convention?
  - Prefix = Module
  - Suffix = Type of table
- Which Module Is Most Likely To Have The Facts?

	A	B
1		
2	**Table Prefix**	**KFS Module**
3		
4	**AP**	Accounts Payable
5	**AR**	Accounts Receivable
6	**CA**	Chart Of Accounts
7	**CB**	Capital Assets
8	**CG**	Contract And Grants
9	**CM**	Capital Assets
10	**CU**	Cornell Enhancement
11	**ER**	Endowment
12	**FP**	Financial Processing
13	**FS**	Financial Processing
14	**GL**	General Ledger
15	**KRN**	Kuali Rice Nervous System
16	**KRS**	Kuali Rice System
17	**LD**	Labor Ledger
18	**PDP**	Pre-Disbursement Processing
19	**PS**	Position
20	**PUR**	Purchasing
21	**QRT**	Quartz
22	**SEC**	Security
23	**SH**	Shared

Research GL (General Labor) Tables

1	TABLE_NAME	COLUMN_NAME	DATA_TYPE	DATA_PRECISIO	DATA_SCA	COLUMN_	IS_PRIMARY_KI
3416	GL_ACCT_BALANCES_HIST_T	UNIV_FISCAL_YR	NUMBER	4	0	1	Y
3417	GL_ACCT_BALANCES_HIST_T	FIN_COA_CD	VARCHAR2			2	Y
3418	GL_ACCT_BALANCES_HIST_T	ACCOUNT_NBR	VARCHAR2			3	Y
3419	GL_ACCT_BALANCES_HIST_T	SUB_ACCT_NBR	VARCHAR2			4	Y
3420	GL_ACCT_BALANCES_HIST_T	FIN_OBJECT_CD	VARCHAR2			5	Y
3421	GL_ACCT_BALANCES_HIST_T	FIN_SUB_OBJ_CD	VARCHAR2			6	Y
3422	GL_ACCT_BALANCES_HIST_T	CURR_BDLN_BAL_AMT	NUMBER	19	2	7	N
3423	GL_ACCT_BALANCES_HIST_T	ACLN_ACTLS_BAL_AMT	NUMBER	19	2	8	N
3424	GL_ACCT_BALANCES_HIST_T	ACLN_ENCUM_BAL_AMT	NUMBER	19	2	9	N
3425	GL_ACCT_BALANCES_T	UNIV_FISCAL_YR	NUMBER	4	0	1	Y
3426	GL_ACCT_BALANCES_T	FIN_COA_CD	VARCHAR2			2	Y
3427	GL_ACCT_BALANCES_T	ACCOUNT_NBR	VARCHAR2			3	Y
3428	GL_ACCT_BALANCES_T	SUB_ACCT_NBR	VARCHAR2			4	Y
3429	GL_ACCT_BALANCES_T	FIN_OBJECT_CD	VARCHAR2			5	Y
3430	GL_ACCT_BALANCES_T	FIN_SUB_OBJ_CD	VARCHAR2			6	Y
3431	GL_ACCT_BALANCES_T	CURR_BDLN_BAL_AMT	NUMBER	19	2	7	N
3432	GL_ACCT_BALANCES_T	ACLN_ACTLS_BAL_AMT	NUMBER	19	2	8	N
3433	GL_ACCT_BALANCES_T	ACLN_ENCUM_BAL_AMT	NUMBER	19	2	9	N
3434	GL_ACCT_BALANCES_T	TIMESTAMP	DATE			10	N
3435	GL_BALANCE_HIST_T	UNIV_FISCAL_YR	NUMBER	4	0	1	Y
3436	GL_BALANCE_HIST_T	FIN_COA_CD	VARCHAR2			2	Y
3437	GL_BALANCE_HIST_T	ACCOUNT_NBR	VARCHAR2			3	Y

Research GL Tables

- How do you find a foundational fact in the GL Tables?
- Remember that facts are NUMBERs.
- Filter by Data Type = NUMBER
- Examine Column Names and see if you can identify a foundational fact.
- What have you learned about the source data?
  - Abbreviation “AMT” = Amount = $$$
  - Data Type NUMBER(19,2) = $$$

Research GL Tables

1	TABLE_NAME	COLUMN_NAME	DATA_TYPE	DATA_PRECISIO	DATA_SCA	COLUMN_
3390	GL_ACCT_BALANCES_HIST_T	CURR_BDLN_BAL_AMT	NUMBER	19	2	7
3391	GL_ACCT_BALANCES_HIST_T	ACLN_ACTLS_BAL_AMT	NUMBER	19	2	8
3392	GL_ACCT_BALANCES_HIST_T	ACLN_ENCUM_BAL_AMT	NUMBER	19	2	9
3399	GL_ACCT_BALANCES_T	CURR_BDLN_BAL_AMT	NUMBER	19	2	7
3400	GL_ACCT_BALANCES_T	ACLN_ACTLS_BAL_AMT	NUMBER	19	2	8
3401	GL_ACCT_BALANCES_T	ACLN_ENCUM_BAL_AMT	NUMBER	19	2	9
3411	GL_BALANCE_HIST_T	ACLN_ANNL_BAL_AMT	NUMBER	19	2	9
3412	GL_BALANCE_HIST_T	FIN_BEG_BAL_LN_AMT	NUMBER	19	2	10
3413	GL_BALANCE_HIST_T	CONTR_GR_BB_AC_AMT	NUMBER	19	2	11
3414	GL_BALANCE_HIST_T	MO1_ACCT_LN_AMT	NUMBER	19	2	12
3415	GL_BALANCE_HIST_T	MO2_ACCT_LN_AMT	NUMBER	19	2	13
3416	GL_BALANCE_HIST_T	MO3_ACCT_LN_AMT	NUMBER	19	2	14
3417	GL_BALANCE_HIST_T	MO4_ACCT_LN_AMT	NUMBER	19	2	15
3418	GL_BALANCE_HIST_T	MO5_ACCT_LN_AMT	NUMBER	19	2	16
3419	GL_BALANCE_HIST_T	MO6_ACCT_LN_AMT	NUMBER	19	2	17
3420	GL_BALANCE_HIST_T	MO7_ACCT_LN_AMT	NUMBER	19	2	18
3421	GL_BALANCE_HIST_T	MO8_ACCT_LN_AMT	NUMBER	19	2	19
3422	GL_BALANCE_HIST_T	MO9_ACCT_LN_AMT	NUMBER	19	2	20
3423	GL_BALANCE_HIST_T	MO10_ACCT_LN_AMT	NUMBER	19	2	21
3424	GL_BALANCE_HIST_T	MO11_ACCT_LN_AMT	NUMBER	19	2	22
3425	GL_BALANCE_HIST_T	MO12_ACCT_LN_AMT	NUMBER	19	2	23
3426	GL_BALANCE_HIST_T	MO13_ACCT_LN_AMT	NUMBER	19	2	24
3435	GL_BALANCE_T	ACLN_ANNL_BAL_AMT	NUMBER	19	2	9
3436	GL_BALANCE_T	FIN_BEG_BAL_LN_AMT	NUMBER	19	2	10
3437	GL_BALANCE_T	CONTR_GR_BB_AC_AMT	NUMBER	19	2	11
3438	GL_BALANCE_T	MO1_ACCT_LN_AMT	NUMBER	19	2	12
3439	GL_BALANCE_T	MO2_ACCT_LN_AMT	NUMBER	19	2	13
3440	GL_BALANCE_T	MO3_ACCT_LN_AMT	NUMBER	19	2	14
3441	GL_BALANCE_T	MO4_ACCT_LN_AMT	NUMBER	19	2	15

Research GL Tables

- We see many potential facts.
- How do you find the lowest level fact?
  - Avoid pre-aggregated data
  - Find atomic transactions

Research GL Tables

1	TABLE_NAME	COLUMN_NAME	DATA_TYPE	DATA_PRECISIO	DATA_SCA	COLUMN_
3447	GL_BALANCE_T	MO10_ACCT_LN_AMT	NUMBER	19	2	21
3448	GL_BALANCE_T	MO11_ACCT_LN_AMT	NUMBER	19	2	22
3449	GL_BALANCE_T	MO12_ACCT_LN_AMT	NUMBER	19	2	23
3450	GL_BALANCE_T	MO13_ACCT_LN_AMT	NUMBER	19	2	24
3485	GL_COR_DOC_T	GL_COR_DEBIT_TOT_AMT	NUMBER	19	2	9
3486	GL_COR_DOC_T	GL_COR_CRDT_TOT_AMT	NUMBER	19	2	10
3490	GL_COR_DOC_T	GL_COR_BDGT_TOT_AMT	NUMBER	19	2	14
3501	GL_ENCUMBRANCE_HIST_T	ACLN_ENCUM_AMT	NUMBER	19	2	11
3502	GL_ENCUMBRANCE_HIST_T	ACLN_ENCUM_CLS_AMT	NUMBER	19	2	12
3515	GL_ENCUMBRANCE_T	ACLN_ENCUM_AMT	NUMBER	19	2	13
3516	GL_ENCUMBRANCE_T	ACLN_ENCUM_CLS_AMT	NUMBER	19	2	14
3526	GL_ENTRY_HIST_T	TRN_LDGR_ENTR_AMT	NUMBER	19	2	8
3542	GL_ENTRY_T	TRN_LDGR_ENTR_AMT	NUMBER	19	2	15
3571	GL_ID_BILL_T	FDOC_IDBIL_ITM_AMT	NUMBER	19	2	17
3587	GL_ORG_RVRSN_CTGRY_AMT_T	ORG_TOT_ACTL_AMT	NUMBER	19	2	7
3588	GL_ORG_RVRSN_CTGRY_AMT_T	ORG_TOT_BDGT_AMT	NUMBER	19	2	8
3589	GL_ORG_RVRSN_CTGRY_AMT_T	ORG_TOT_ENCUM_AMT	NUMBER	19	2	9
3590	GL_ORG_RVRSN_CTGRY_AMT_T	ORG_TOT_CF_AMT	NUMBER	19	2	10
3591	GL_ORG_RVRSN_CTGRY_AMT_T	ORG_TOT_AVAIL_AMT	NUMBER	19	2	11
3597	GL_ORG_RVRSN_UNIT_WRK_T	ORG_TOT_RVRSN_AMT	NUMBER	19	2	6
3598	GL_ORG_RVRSN_UNIT_WRK_T	ORG_TOT_CF_AMT	NUMBER	19	2	7
3599	GL_ORG_RVRSN_UNIT_WRK_T	ORG_TOT_AVAIL_AMT	NUMBER	19	2	8
3600	GL_ORG_RVRSN_UNIT_WRK_T	ORG_TOT_CSH_AMT	NUMBER	19	2	9
3604	GL_PENDING_BALANCES_MT	APPR_BUDGET	NUMBER	19	2	4
3605	GL_PENDING_BALANCES_MT	APPR_ACTUAL	NUMBER	19	2	5
3606	GL_PENDING_BALANCES_MT	APPR_ENCUM	NUMBER	19	2	6
3607	GL_PENDING_BALANCES_MT	PEND_BUDGET	NUMBER	19	2	7
3608	GL_PENDING_BALANCES_MT	PEND_ACTUAL	NUMBER	19	2	8
3609	GL_PENDING_BALANCES_MT	PEND_ENCUM	NUMBER	19	2	9
3625	GL_PENDING_ENTRY_MT	TRN_LDGR_ENTR_AMT	NUMBER	19	2	16
3656	GL_PENDING_ENTRY_T	TRN_LDGR_ENTR_AMT	NUMBER	19	2	16
3687	GL_REVERSAL_T	TRN_LDGR_ENTR_AMT	NUMBER	19	2	16

The Foundational Fact!

- GL_ENTRY_T . TRN_LDGR_ENTR_AMT !!!

- How do you test your hypothesis?
  - Use the application to create a test transaction and see where the data goes (Amount = 12345.67).
  - Does it land in that column?
  - Confirm your finding with SME.
  - Continue your analysis to confirm this is a foundational fact at the lowest grain.

The Hunt for Dimensions

Now that you have chosen a fact, how do you find the associated dimensions?

Dimensions categorize and describe facts in ways that support meaningful answers to business questions.

What can you slice-and-dice this fact by?

Use the primary keys of your fact to lead you to the dimensions.

The Hunt for Dimensions

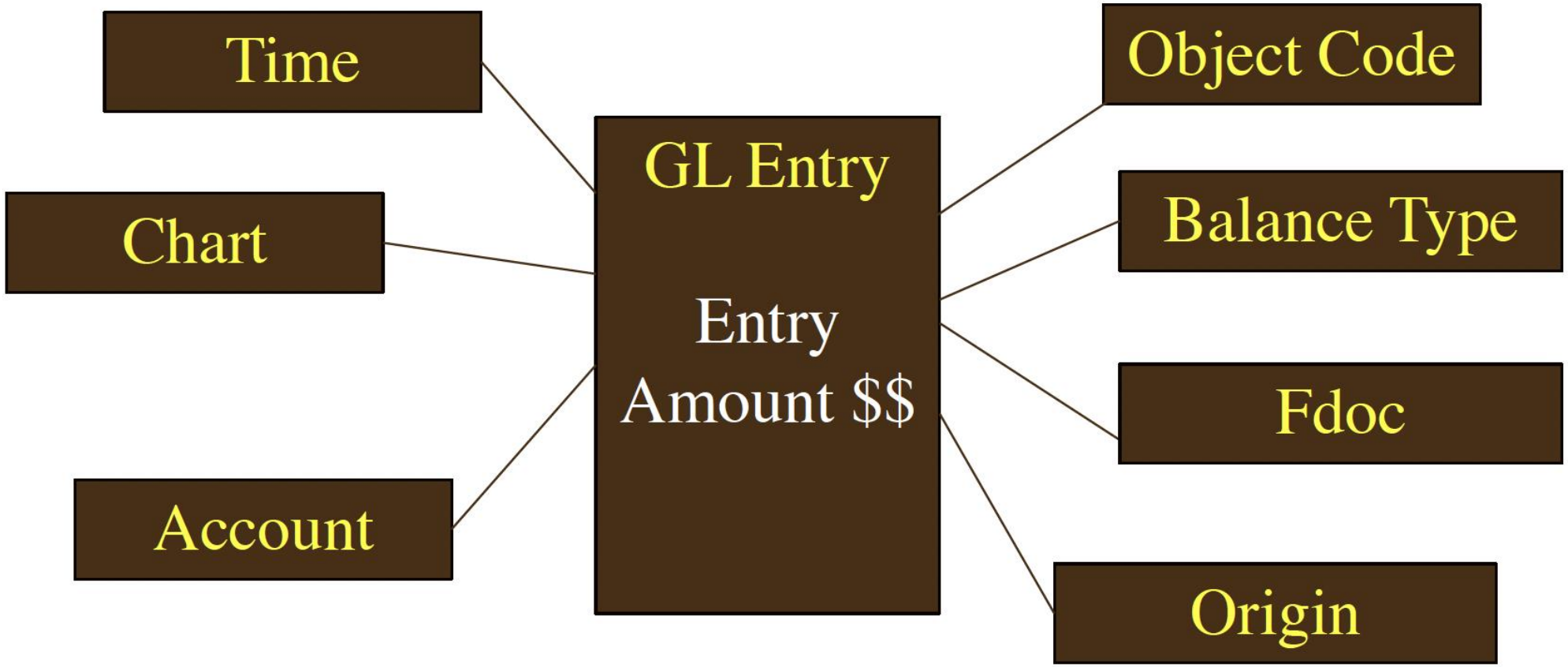
TABLE_NAME	COLUMN_NAME	DATA_TYPE	DATA_PRECISIO	DATA_SCA	COLUMN_	IS_PRIMARY_KI
GL_ENTRY_T	UNIV_FISCAL_YR	NUMBER	4	0	1	Y
GL_ENTRY_T	FIN_COA_CD	VARCHAR2			2	Y
GL_ENTRY_T	ACCOUNT_NBR	VARCHAR2			3	Y
GL_ENTRY_T	SUB_ACCT_NBR	VARCHAR2			4	Y
GL_ENTRY_T	FIN_OBJECT_CD	VARCHAR2			5	Y
GL_ENTRY_T	FIN_SUB_OBJ_CD	VARCHAR2			6	Y
GL_ENTRY_T	FIN_BALANCE_TYP_CD	VARCHAR2			7	Y
GL_ENTRY_T	FIN_OBJ_TYP_CD	VARCHAR2			8	Y
GL_ENTRY_T	UNIV_FISCAL_PRD_CD	VARCHAR2			9	Y
GL_ENTRY_T	FDOC_TYP_CD	VARCHAR2			10	Y
GL_ENTRY_T	FS_ORIGIN_CD	VARCHAR2			11	Y
GL_ENTRY_T	FDOC_NBR	VARCHAR2			12	Y
GL_ENTRY_T	TRN_ENTR_SEQ_NBR	NUMBER	5	0	13	Y
GL_ENTRY_T	TRN_LDGR_ENTR_DESC	VARCHAR2			14	N
GL_ENTRY_T	TRN_LDGR_ENTR_AMT	NUMBER	19	2	15	N
GL_ENTRY_T	TRN_DEBIT_CRDT_CD	VARCHAR2			16	N
GL_ENTRY_T	TRANSACTION_DT	DATE			17	N
GL_ENTRY_T	ORG_DOC_NBR	VARCHAR2			18	N
GL_ENTRY_T	PROJECT_CD	VARCHAR2			19	N
GL_ENTRY_T	ORG_REFERENCE_ID	VARCHAR2			20	N
GL_ENTRY_T	FDOC_REF_TYP_CD	VARCHAR2			21	N
GL_ENTRY_T	FS_REF_ORIGIN_CD	VARCHAR2			22	N
GL_ENTRY_T	FDOC_REF_NBR	VARCHAR2			23	N
GL_ENTRY_T	FDOC_REVERSAL_DT	DATE			24	N
GL_ENTRY_T	TRN_ENCUM_UPDT_CD	VARCHAR2			25	N
GL_ENTRY_T	TRN_POST_DT	DATE			26	N
GL_ENTRY_T	TIMESTAMP	DATE			27	N

The Hunt for Dimensions

Likely Dimensions:

- Time
- Chart
- Account
- Object Code
- Balance Type
- Fdoc
- Origin

Conceptual Model



Research the Remaining Dimensions

- Continue and research each of the potential dimensions
  - Chart
  - Fund
  - Function
  - Object Code
  - Etc.

Now What?

- Do we stop here and assume a top-down approach?
- Or do we go further and actually create a physical model?
  - How hard is it to create physical tables?
  - Do I have to use an ETL tool?
  - Won't the risk of getting it wrong be too great?

Prototype the Model

- Use SQL or WhereScape to prototype the model.
- Create the dimensions.
- Create the fact
- Test the prototype

Prototype the Dimensions

- Need naming conventions
  - Table names
    - Prefix
    - Suffix
    - Abbreviations
  - Column names
  - Column order

Prototype the Fact

- Usually a 1:1 copy of the source table with the SIDs added and columns renamed.
- SIDs at the top
- Fact(s) at the bottom
- Natural keys in the middle

Test Your Prototype Model

- Test the prototype using SQL and/or a BI Tool
- Show the prototype in the BI Tool to users - allow users to kick the tires.
- Does it answer the business questions that they want to ask?
- Helps users refine requirements
- Helps users learn about the data
- Can help validate new data

Test Your Prototype Model

- Triggers thoughts that probably would not have come up from conventional requirements gathering.
- What is the gap between the prototype and Top-Down requirements?
- Enhance design based upon Top-Down requirements.
  - Calculated/Derived Columns
  - YTD, YAGO, MAGO etc …

Benefits of Bottom-Up

- Top-Down Requirements Will Enhance The Data Model - Usually in the form of calculated columns
- Your prototype models can help the users validate that they converted legacy data correctly, that they configured the application correctly well before they go-live.

Cons of Bottom-Up

- Does not consider the Enterprise as a whole.
- You may get it wrong.
- It is harder than Top Down - Requires you dig into the application and figure out the data usually before the end user even understands it.